Grace Ojemerenvhie Alele

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RESEARCH INTEREST

Multimodal AI for Reproductive Health
Disease Discovery in Low-Resource Settings
Clinical AI Safety
Medical Imaging

EDUCATION

Certificate — **The Johns Hopkins University, Coursera** — **In - view**
Genomic Data Science Specialization

BSc (Honours) — **Ambrose Alli University, Nigeria Computer Science** — **2021 - 2025**
- Thesis: A Lightweight Transcriptomic Pipeline for Molecular Subtype Discovery in Postpartum Psychosis
- Advisor: Prof S.E. Nnebe
- Rank: Top 5% of Class

GRE — 331/340 (Quant: 164, Verbal: 167)

RESEARCH EXPERIENCE

2026 – Present
Machine Learning Collective (MLC)
Independent AI/ML Researcher
- Developing uncertainty-aware multimodal transformer for endometriosis diagnosis, integrating structured patient symptom data, virtual contrast MRI synthesis with lesion-aware segmentation to improve safety and clinical reliability. Investigating geographic generalization of the pipeline across international patient populations, with a long-term focus on extending diagnostic infrastructure to Africa clinical contexts.
- Investigating geographic representation bias in tabular ML benchmarks and developing an African dataset suite to assess performance and fairness gaps in classical models vs LLMs.
- Answering The Question: Can Mamba-based SSMs (and Mamba + CNN/ViT hybrids) model both local detail and global context in 3D brain MRI segmentation with lower memory and compute cost than ViT-heavy solutions, while matching or exceeding clinical segmentation metrics for state of the art models?
- Developing a shift-aware safety metrics (Risk-Coverage Shift, AURC) to audit confidence calibration in clinical LLMs when transferred from Western (USMLE) to African (AfriMed) contexts (geographic shift).

2025 – Present
AI for food Allergies (Hugging Face Science)
Research Assistant
- Engineered the development of foundational tools for scalable integrative analysis of complex, longitudinal microbiome data using the Diabimmune cohort dataset (1,000+ sample).
- I co-led the model-development phase of the project, transforming prototypes into robust, reproducible systems by refactoring and modularizing the codebase.
- Engineered a Python API client with retry logic for querying microbial genomic databases and developed a rule-based labeling pipeline to infer allergy phenotypes with 85% cross-validation accuracy.
- Diagnosed and fixed a foundational bug in the embedding methodology by implementing proper attention masking, eliminating batch-size dependency and ensuring consistent results. View

2025
Ambrose Alli University
Honour Thesis: NeuroRepro-Seq: A Lightweight Transcriptomic Pipeline for Molecular Subtype Discovery in Postpartum Psychosis
- Developed a computationally efficient RNA-seq pipeline for identifying molecular disease subtyping in postpartum psychosis.
- Incorporated bootstrap ARI stability validation, Mann-Whitney marker identification, and literature-curated PPP

gene signatures across neuroinflammatory, HPA/hormonal, and dopaminergic pathways
- Optimised for small clinical cohorts (n<25) and low-resource environments (<400 MB RAM). View

2025 **Ambrose Alli University**

Undergraduate Group Work –Title: Development and Design of A Blockchain Solar Energy Trading Platform

- Co-led the design of a decentralized peer-to-peer (P2P) solar energy trading platform using blockchain technology for secure, transparent transactions.
- Designed the system architecture and implemented solidity for automated energy trading and settlement.
- Integrated IoT-enabled smart meter concepts for real-time monitoring of energy production and consumption.
- Conducted literature review and technical analysis on scalability, security, and regulatory constraints in blockchain energy markets; co-authored the final technical report and presentation. View

2023 – 2024 **Afe Babalola University (ABUAD)**

Undergraduate Research Internship

- Addressed the computationally expensive hyperparameter tuning for multimodal ML models segmenting Glioblastoma (aggressive brain tumors) in MRI scans by developing a bayesian optimization pipeline to combine high-dimensional MRI feature sets into a unified representation, automating the tuning process.
- Established an efficient optimization framework that is scalable to other complex biological datasets.
- I achieved a **40% reduction in GPU compute hours** and a **60% reduction in tuning trials**, while maintaining a clinically robust segmentation accuracy (**Dice Score of 0.89**). View

PUBLICATIONS, WORKSHOPS, MANUSCRIPTS AND POSTERS

4. Oko-odion, T. U., Tiwari, Asama, T., A., **Ojemerenvhie, G.A**, &. Amin, E. (2026). "Selective Risk Inflation under Geographic Distribution Shift in Small Medical LLMs." **[ACL (in review)]**
3. Akintayo, T.A., ..., **Ojemerenvhie, G.A.,** et al. (2024). "Exploring the Possibilities of AI in Medical Settings: How Artificial Intelligence Can Transform Healthcare and Hospital Operations." International Journal of Advances in Engineering and Management.
2. Akintayo, T.A., ..., **Ojemerenvhie, G.A.,** et al. (2024). "Transforming Data Analytics with AI for Informed Decision-Making." International Journal of Education, Management, and Technology, 2(3), 196-215.
1. Okereke, R.O., **Ojemerenvhie, G.A.,** et al. (2024). "The Cloud Security Revolution: Unlocking the Potential of Al and Machine Learning to Stay Ahead of Threats." Asian Journal of Science, Technology, Engineering, and Art, 2(5), 735-743.

TEACHING EXPERIENCE

2025 **Assistant Course Instructor – CSC308: Operations Research**

Department of Computer Science, Ambrose Alli University

- Delivered in-class lectures and facilitated active learning experiences, employing pedagogical strategies that encouraged critical thinking, student participation, and collaborative problem-solving.
- Contributed to the revision of the course textbook, critically reviewing and refining explanations of core programming concepts to enhance clarity and pedagogical effectiveness, while updating examples, exercises, and code snippets to align with current technological trends and real-world applications.
- Designed and administered formative assessments including online polls, assignments, and quizzes and provided detailed and structured feedback on correctness, algorithmic efficiency, and code readability to both students and course instructors.
- Graded assignments and supervised tests and examinations, ensuring academic integrity and consistent application of evaluation criteria.
- Delivered 8+ hours per week of structured academic support across laboratory sessions and office hours, serving CS students across multiple course sections with consistent, high-quality instructional assistance.

2024 - 2025 **Lab Assistant – CSC305: Artificial Intelligence**

Department of Computer Science, Ambrose Alli University

- Bridged the gap between theoretical coursework and real-world implementation by conducting practical workshops and hands-on training sessions covering core AI concepts including search algorithms, knowledge representation, machine learning fundamentals, and intelligent agent design.

- Guided students through the configuration of development environments and provided targeted debugging and algorithmic support in Python and Git-based workflows, resolving version control, dependency, and runtime issues encountered during live lab sessions.
- Facilitated project-based workshops where students designed, trained, and evaluated basic AI models, applying concepts such as decision trees, rule-based systems, and introductory neural networks to practical problems.
- Supported course infrastructure management via GitHub Classroom and automated grading tools, ensuring seamless assignment distribution, submission tracking, and evaluation of programming deliverables.
- Cultivated systematic thinking and sound software engineering practices by mentoring student teams through the full lifecycle of AI-focused project assignments from problem scoping and algorithm selection through implementation, testing, and final evaluation.

2023 - 2025 **Assistant Course Instructor – CSC211: Data Structure and Algorithm**
Department of Computer Science, Ambrose Alli University

- Instructed students on foundational and advanced Data Structures and Algorithm concepts, contextualizing abstract theories through real-world examples, hands-on coding exercises, and annotated code snippets to reinforce comprehension and practical applicability.
- Cultivated an intellectually engaging classroom environment by designing interactive learning experiences that promoted active discourse, peer collaboration, and independent analytical reasoning.
- Evaluated student progress through a multi-modal assessment approach encompassing online polls, assignments, and quizzes and delivered comprehensive feedback to course instructors, with emphasis on solution correctness, computational efficiency, and code readability.

RELEVANT PROFESSIONAL EXPERIENCE

2022 – Present **Edupoint Ltd.** – Robotics and Coding Instructor (Part-Time)

- I teach primary and secondary school students logic, sequencing, problem-solving, through coding and unplugged activities using Scratch, Python, C++ and Blockly.
- Apply algorithms to physical components, integrating mechanics, electronics, sensors, and systems thinking.
- Assessed student learning through interactive activities, assignments and quizzes

2019 – 2020 **BrighterDays CodeLab** – Software Developer Trainee

- I developed Flutter apps with Material Design, focusing on UI/UX and code reusability and integrated the apps with Facebook and Google APIs for authentication and maps.
- I collaborated with designers and translated project requirements into robust implementations.
- I was introduced to Machine Learning, Deep Learning and Artificial Intelligence basics by Mr Peter Oti.

SKILLS AND TOOLS

Research Computing:	Machine Learning, Deep Learning, Model Evaluation & Benchmark Design, Data Preprocessing & Feature Engineering, EDA, Experimental Methodology and Statistical Testing
Machine Learning:	Python (PyTorch, TensorFlow, NumPy, Pandas and Scikit-learn), R, SQL, Bash/Shell Scripting, Statistical Analysis & Data Visualization, Git/GitHub and Docker Containerization.
Biomedical Data Analysis:	Scanpy, Seurat, and DESeq2, Data processing (SAMtools and BEDtools)
Software Engineering:	Dart/Flutter, Django, Tkinter, FastAPI and Web-based Repository Development.

MENTORING ACTIVITIES

Mentored junior undergraduates in programming, career development, and community leadership, including:

2024 – Present	**Seun Olawunmi,** Django Girls Bootcamp AAU
2023 – Present	**Soribe Stella**, TEDx AAU Ekpoma
2023 – Present	**Destiny Ose & Christabel Christopher,** Google Developer Student Club AAU
2022 – Present	**Daniel Path**, Data Science Network Nigeria AAU

SERVICE

2021 – 2025 Founder, GEEKS Tutorial Group

2024 – 2025	Secretary General, Nigerian Association of Computing Students (NACOS), AAU Chapter
2024	Co-organizer & Mentor, Django Girls Bootcamp AAU
2023/2024	Lead Technical Coordinator, TEDxAAU Ekpoma
2023 – 2024	Google Developer Student Clubs (GDSC) Lead
2023 – 2024	Leadership & Resource Innovation (LRI) Director: Student Fellowship
2023	Data Clark Volunteer, Centre for Reproductive Health Research and Innovation (CRHRI)
2022	Campus Lead and Community Manager, Data Science Network Nigeria

SCHOLARSHIP OFFERS AND HONOURS (TOTAL VALUE: $80,600)

Graduate Merit Scholarship (Emory University), 2026: Awarded a 20% tuition fellowship for each term until graduation for the MS in Computer Science, valued at approximately $18,500 total (Declined).

SCI Scholarship for Excellence in Master's Studies (University of Pittsburgh), 2026: Awarded a $7650 merit-based scholarship ($2,550 per term for three terms) in recognition of strong academic qualification and potential for the M.S. in Intelligent Systems program. (Declined).

CIT Non-Resident Graduate Tuition Scholarship, Global Graduate Merit Scholarship (University of Michigan), 2025: Awarded a total of $10,400 annually based on outstanding academic qualifications (Declined).

Graduate Dearborn Scholarship (University of Michigan), 2025: Awarded a total of $10,000 annually based on outstanding academic qualifications (Declined).

DataCamp/Data Science Network (DSN) Ambassador Scholarship 2023: Awarded a $150 full-tuition scholarship for advanced technical training in Tech.

Unicaf Undergraduate Scholarship (Unicaf), 2022: Awarded an approximately £9,000 merit-based scholarship covering 80% of tuition (Declined).

BrighterDays CodeLab Tech Scholar, 2019: Awarded a $1,500 full-tuition scholarship for advanced technical training in software development.

PROFESSIONAL AFFILATIONS & MEMBERSHIP

2025 – Present	International Society for Data Science and Analytics (ISDSA)
2024 – Present	The Endometriosis Support Group Nigeria (ESGN)
2024 – Present	Women in Data (WiD)
2021 – Present	Data Science Network (DSN)

PRESENTATION & TRAINING

Presentation

- **Poster Presentation**, ABUAD-CMHS Scientific Conference
    - "Development of smart footwear for continuous health monitoring," Afe Babalola University, 2025.
    - "An infant incubator design for intermittent access to electricity," Afe Babalola University, 2025.
- **Honours Thesis Defense**, "Design of a Web-Based Academic Repository for Data Integrity," Ambrose Alli University, 2025
- **Seminar Presentation**, "Application of Machine Learning Techniques for Model Parameterization in Biomedical Imaging," Ambrose Alli University, 2024.

Training

- Facilitator: NACOS Tech Bootcamp AAU, 2025
- Training Facilitator: "Django Girls BootCamp AAU" Django Girls Nigeria, 2024
- Organizer/Facilitator: "Solution Challenge: Ideathon" Google Developer Student Club, 2023

LANGUAGES

English: Native Language, Distinguished levels in Listening, Speaking, Reading, and Writing